

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

School of Computing — Department of Computer Science & Engineering

Student Name: Sautrik Roy	Reg No: RA2511003010052
Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit II | Session 1 | SLO 2: Constructors, this Keyword & Methods (Code: 2012)

Q: Activity: Design a Box class that implements Default, Parameterized, and Copy Constructors. Include a method to compute volume and demonstrate the use of the 'this' keyword to resolve shadowing.

Ans: Java Implementation:

```
class Box {
    double width, height, depth;

    // 1. Default Constructor
    Box() {
        this.width = 1.0;
        this.height = 1.0;
        this.depth = 1.0;
    }

    // 2. Parameterized Constructor with 'this' reference
    Box(double width, double height, double depth) {
        this.width = width;
        this.height = height;
        this.depth = depth;
    }

    // 3. Copy Constructor
    Box(Box other) {
        this.width = other.width;
        this.height = other.height;
        this.depth = other.depth;
    }

    double volume() {
        return width * height * depth;
    }
}

public class BoxDemo {
    public static void main(String[] args) {
        Box b1 = new Box();
        Box b2 = new Box(10.0, 5.0, 2.0);
        Box b3 = new Box(b2);

        System.out.println("Default Box Volume: " + b1.volume());
        System.out.println("Parameterized Box Volume: " + b2.volume());
        System.out.println("Copied Box Volume: " + b3.volume());
    }
}
```